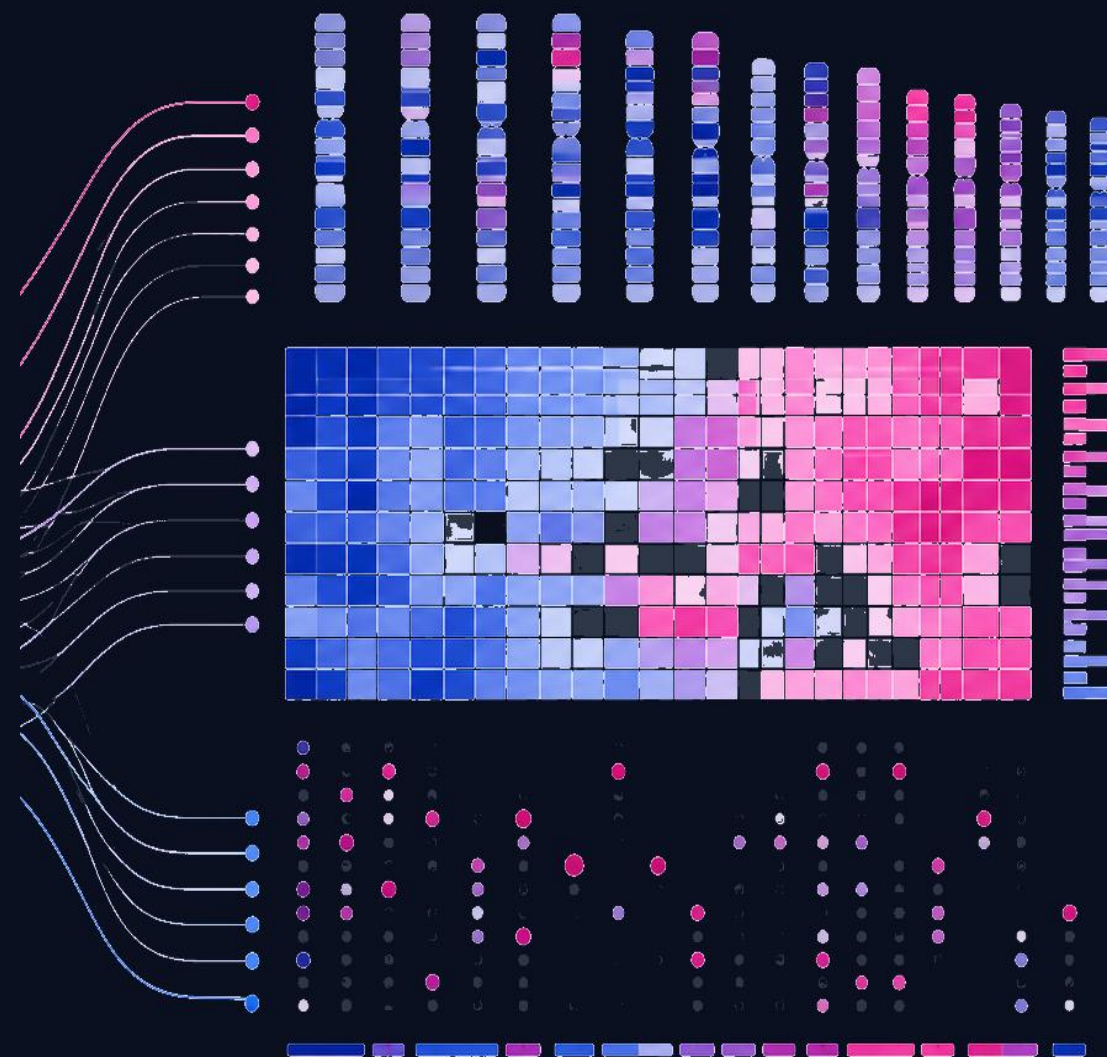


WEEK 2 MEETING

From cBioPortal Exploration to Research Questions

Molecular Profiling of Breast Cancer Subtypes

Question design



Two tabs that can cause confusion

The interface matters: the two views answer different questions.

Genomic Alterations

Use this to find genes that differ between groups.

- Ranked comparison table
- q-values / significance
- alteration frequency by group
- best source for candidate gene list

Question it answers: Which genes are most different between BRCA_Basal and BRCA_LumA?

Mutations Beta

Use this to inspect one gene's mutation pattern.

- lollipop mutation plot
- mutation location along protein
- missense, truncating, splice, in-frame
- not a q-value-ranked table

Question it answers: Where are TP53 or PIK3CA mutations located, and what types are they?

Types of Questions That Can Be Raised

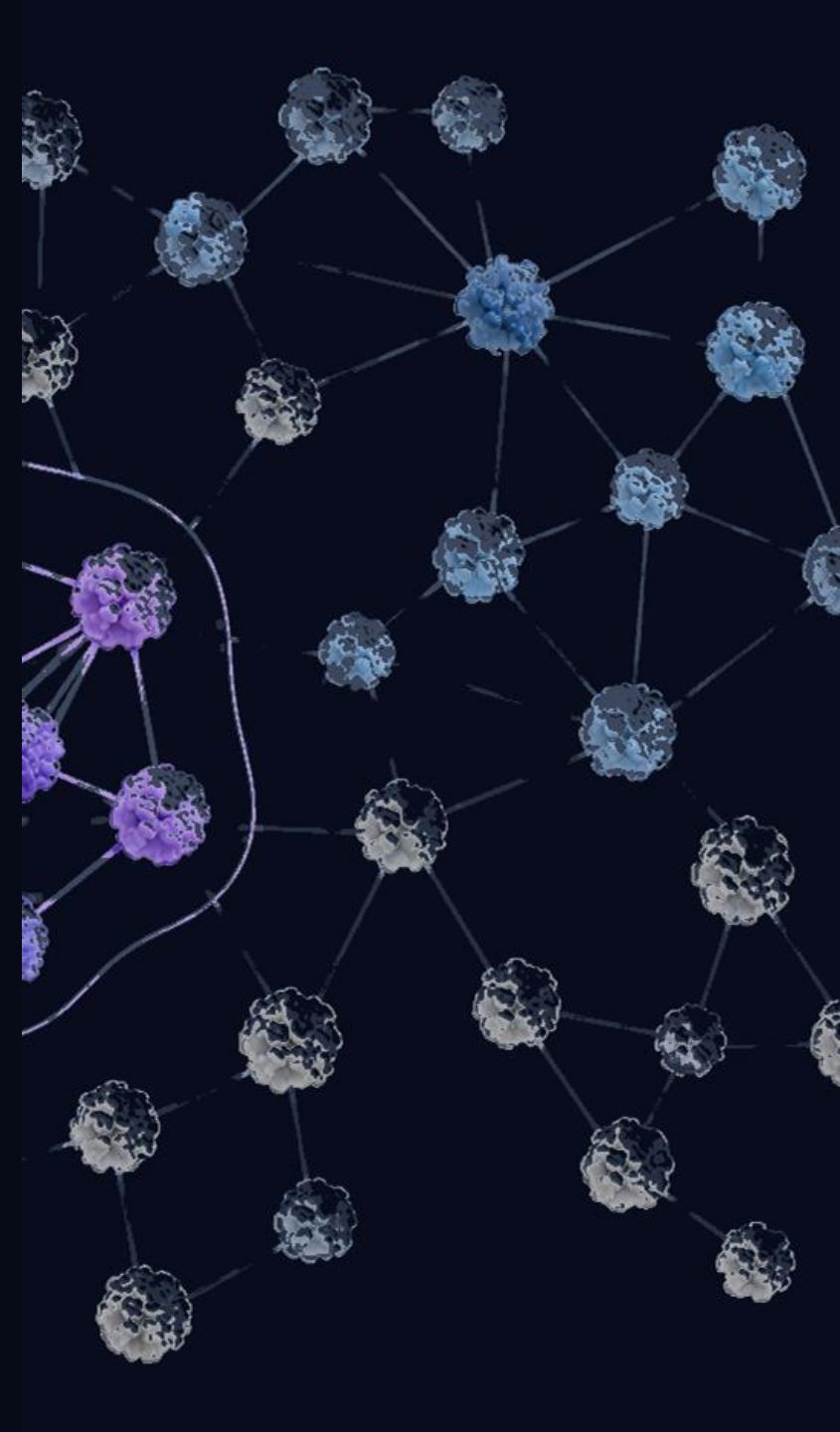
Use the figure first; let the question follow the evidence.

“We compared _____ with _____. We observed that _____ was more common in _____. The figure that supports this is _____. One question this raised was _____.”

- The raised question should be specific, comparative, and answerable with cBioPortal.
- A good question asks about a gene, alteration type, subtype, or patient outcome.
- Avoid questions that require proof of mechanism or drug response unless you frame them as future experiments.

cBioPortal outputs to use

Subtype chart • Genomic Alterations • OncoPrint • Lollipop plot • Survival curve



Reporting

A finding must be tied to a specific figure, not just a general impression.

Use this sentence framework:

“We compared _____ with _____. We observed that _____ was more common in _____. The figure that supports this is _____. One question this raised was _____.”

Example:

“We compared Basal-like with Luminal A tumors. We observed that TP53 mutations were more common in Basal-like tumors. The figure that supports this is the Genomic Alterations comparison. One question this raised was whether TP53 mutations are also associated with different survival outcomes.”

Acceptable evidence

- Subtype chart
- Genomic Alterations table
- Mutations Beta lollipop plot
- OncoPrint
- survival curve

Not enough

- “It looked interesting”
- a gene list without comparison
- a screenshot with no explanation
- a survival curve interpreted as proof of causation

Subtype-Difference Questions

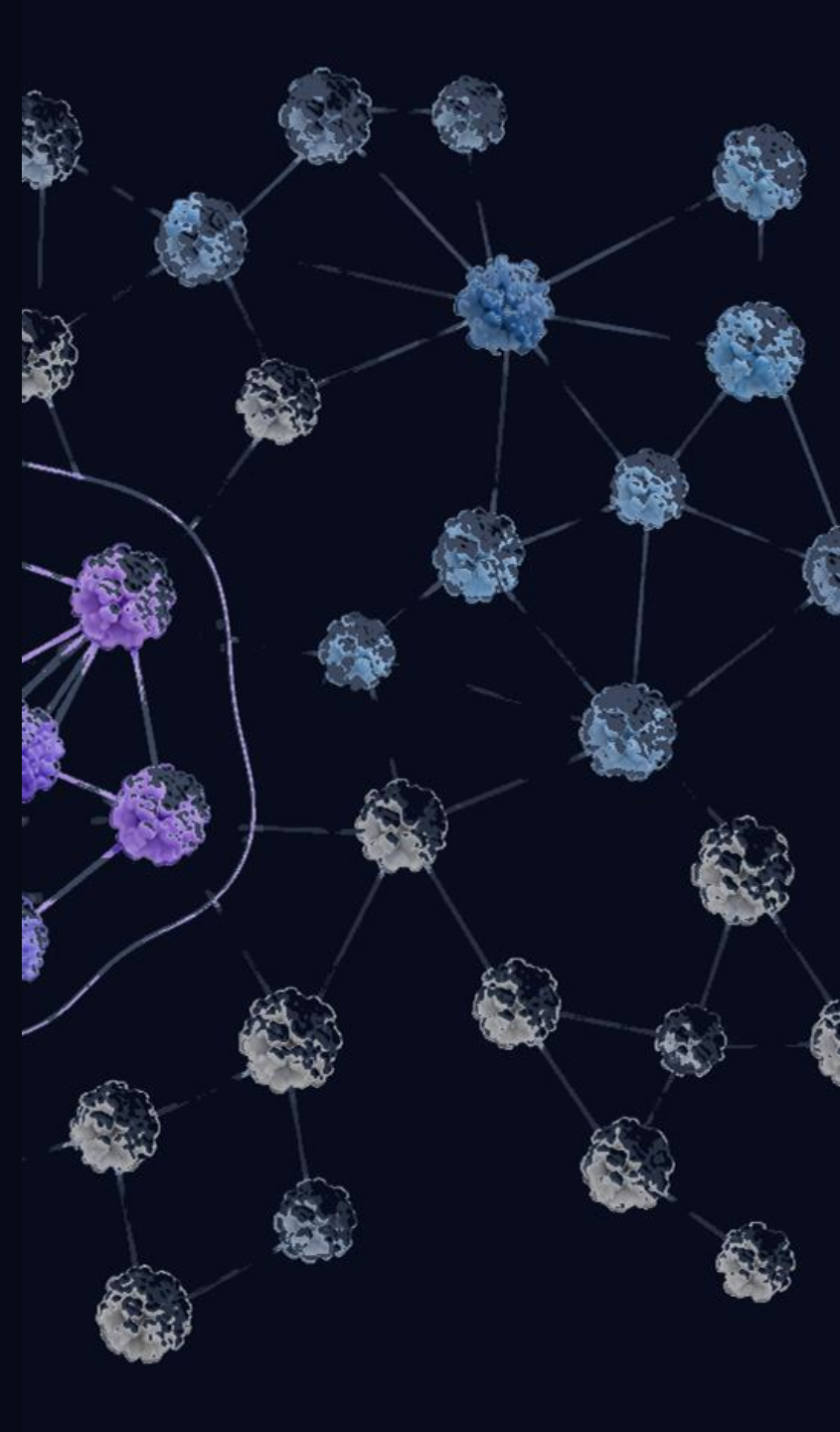
Ask what distinguishes one molecular subtype from another.

Focus: Basal-like versus Luminal A genomic differences

- Why is TP53 altered more often in Basal-like tumors than in Luminal A tumors?
- Which other genes distinguish Basal-like from Luminal A breast cancer?
- Are copy-number alterations also different between these subtypes?
- Is the observed difference limited to one gene or part of a broader genomic pattern?

[cBioPortal outputs to use](#)

Subtype chart • Genomic Alterations comparison • OncoPrint



Candidate-Target Questions

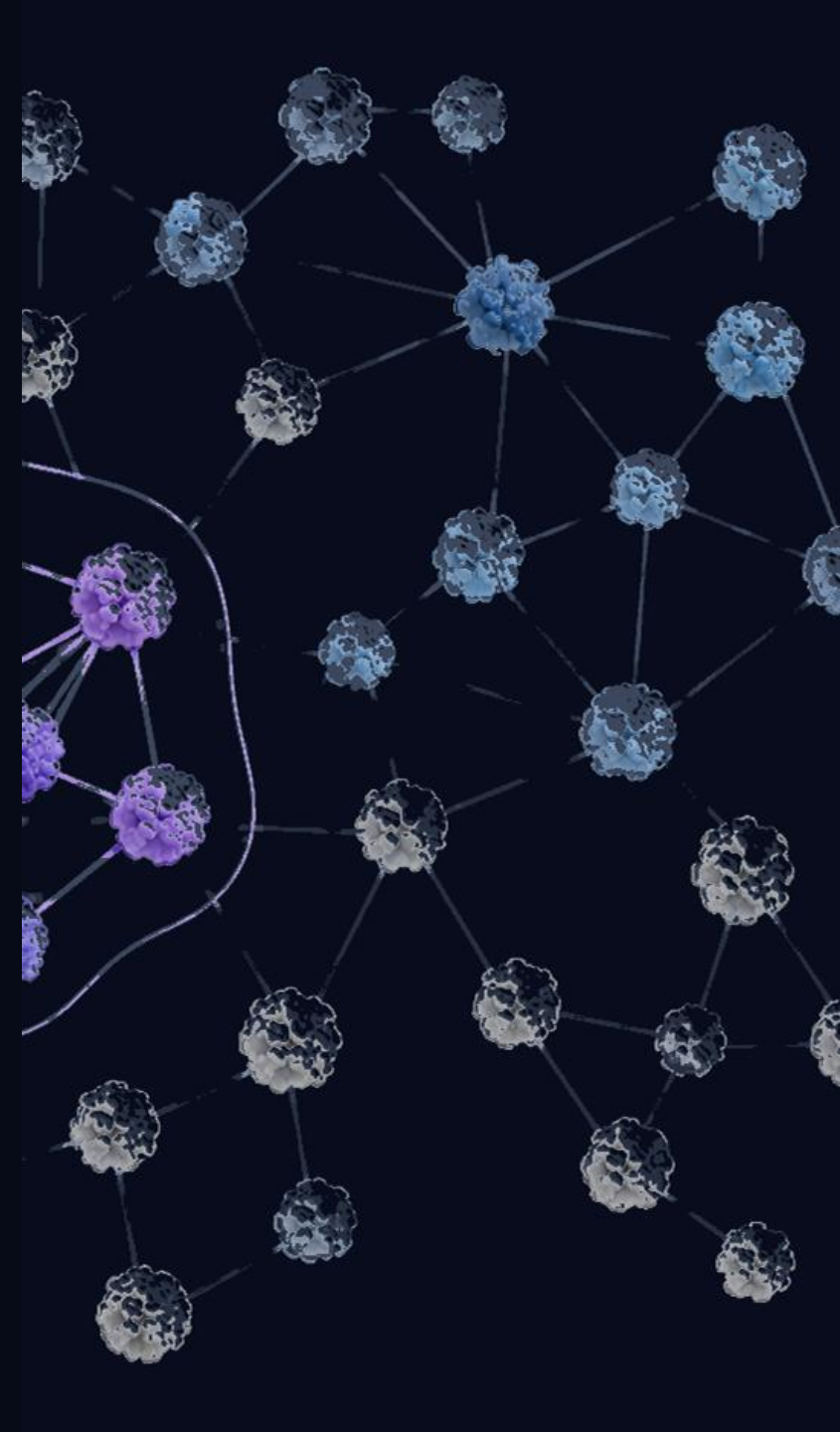
Ask whether a gene is worth prioritizing for follow-up, not whether it is “proven.”

Focus: evidence that a gene may be subtype-specific and clinically relevant

- Is this gene altered often enough to be considered a plausible therapeutic target?
- Is the alteration concentrated in one subtype?
- Does alteration of this gene associate with worse survival?
- Is the gene usually mutated, amplified, or deleted?
- Does the alteration pattern suggest activation of an oncogene or loss of a tumor suppressor?

cBioPortal outputs to use

Genomic Alterations • OncoPrint • Survival curve



Mutation-Pattern Questions

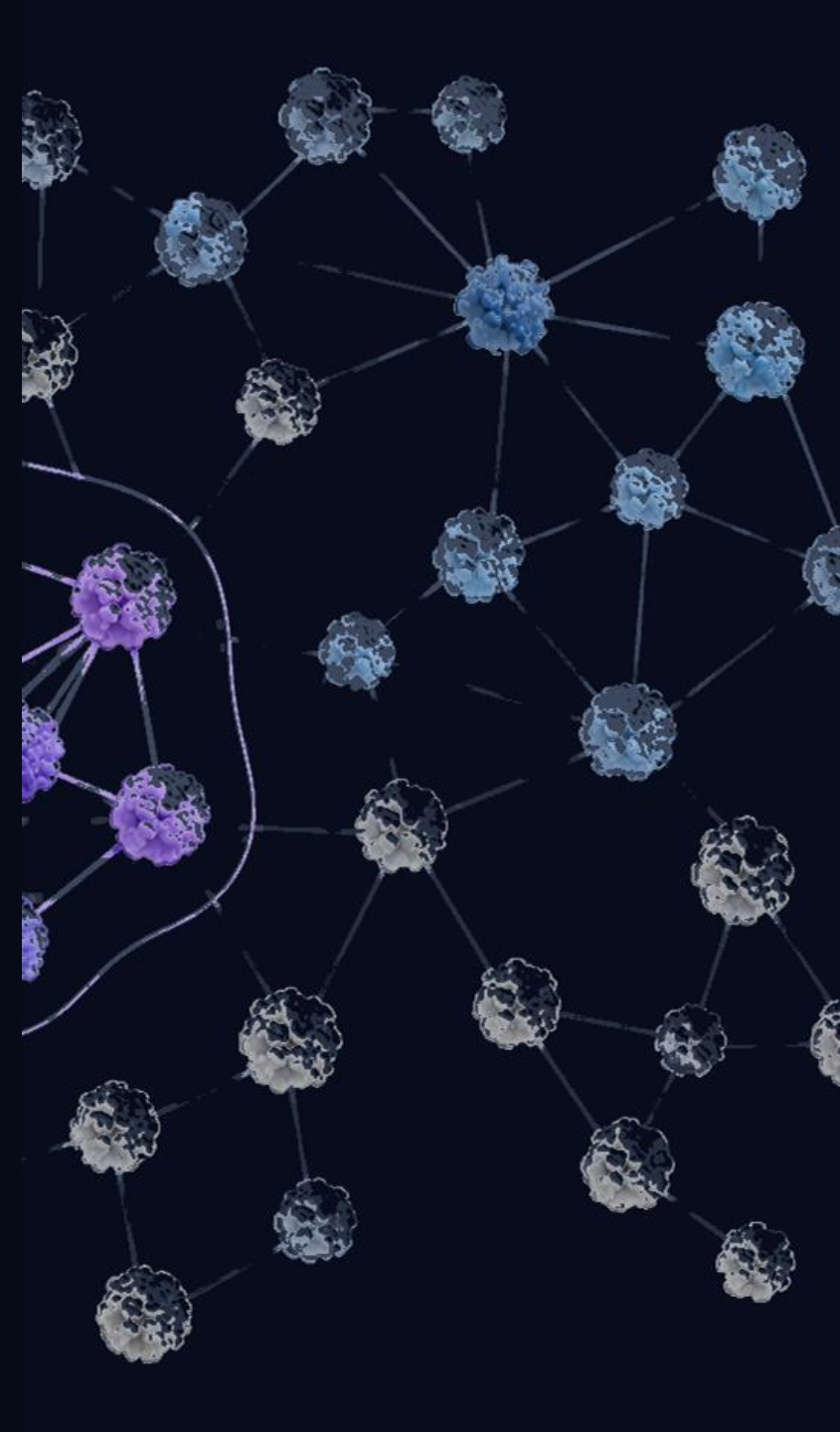
Ask whether mutation frequency, type, or protein location differs by subtype.

Focus: where mutations occur and what kind of mutations they are

- Do the two subtypes differ only in mutation frequency, or also in mutation type?
- Are mutations clustered at specific positions in the protein?
- Are there recurrent hotspot mutations?
- Are truncating mutations more common in one subtype?
- Does one subtype show a broader range of mutation locations?

cBioPortal outputs to use

Mutations Beta lollipop plot • Genomic Alterations • OncoPrint



Survival Questions

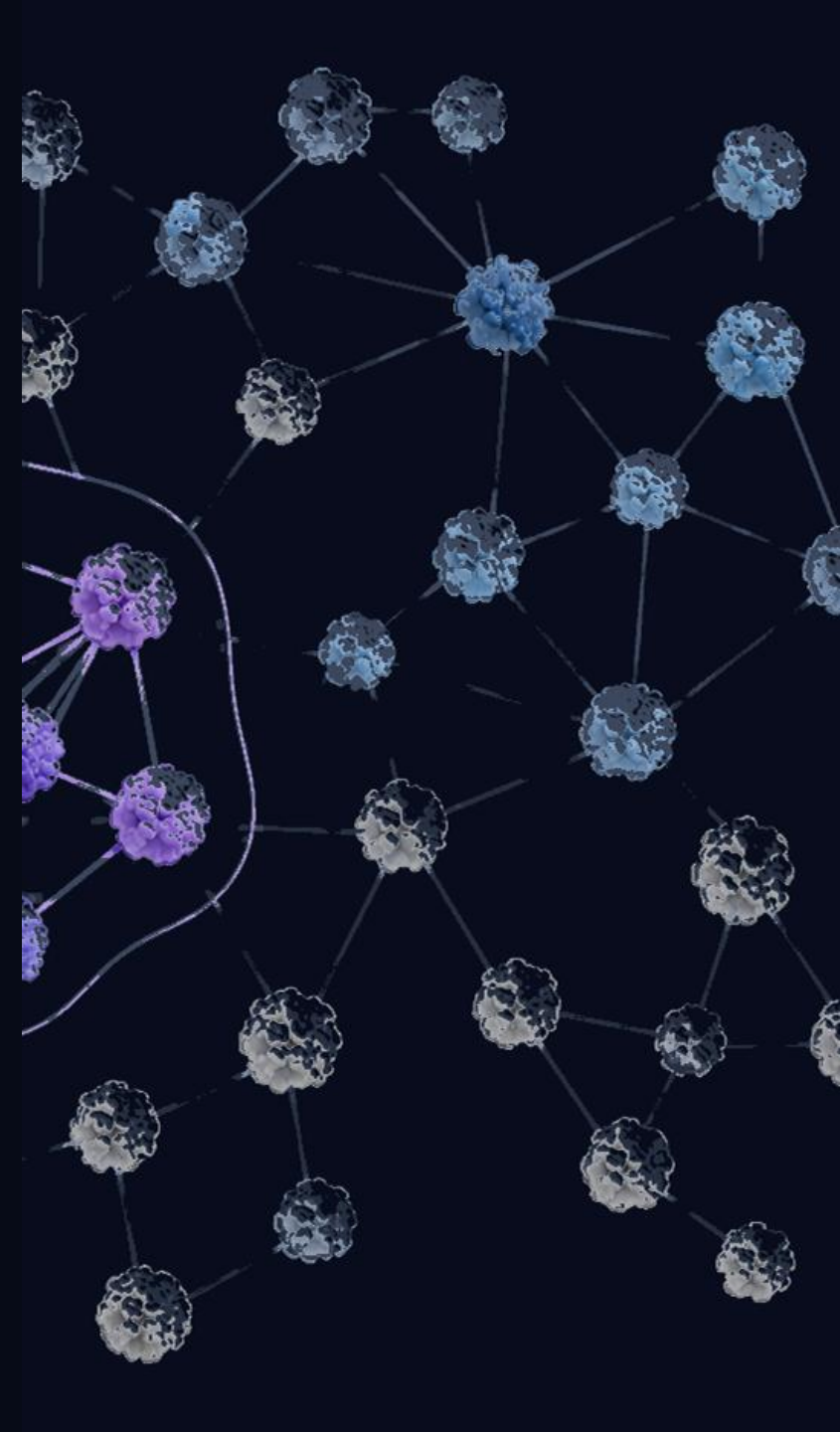
Ask about association with outcome, not causation.

Focus: whether groups have different overall survival patterns

- Do patients with the alteration have different overall survival than patients without it?
- Do Basal-like and Luminal A tumors differ in survival?
- Is the survival difference statistically significant?
- Could other factors, such as stage or treatment, explain the difference?
- Is the survival association strong enough to justify further study?

cBioPortal outputs to use

Survival curve • log-rank p-value • OncoPrint-defined groups



OncoPrint Questions

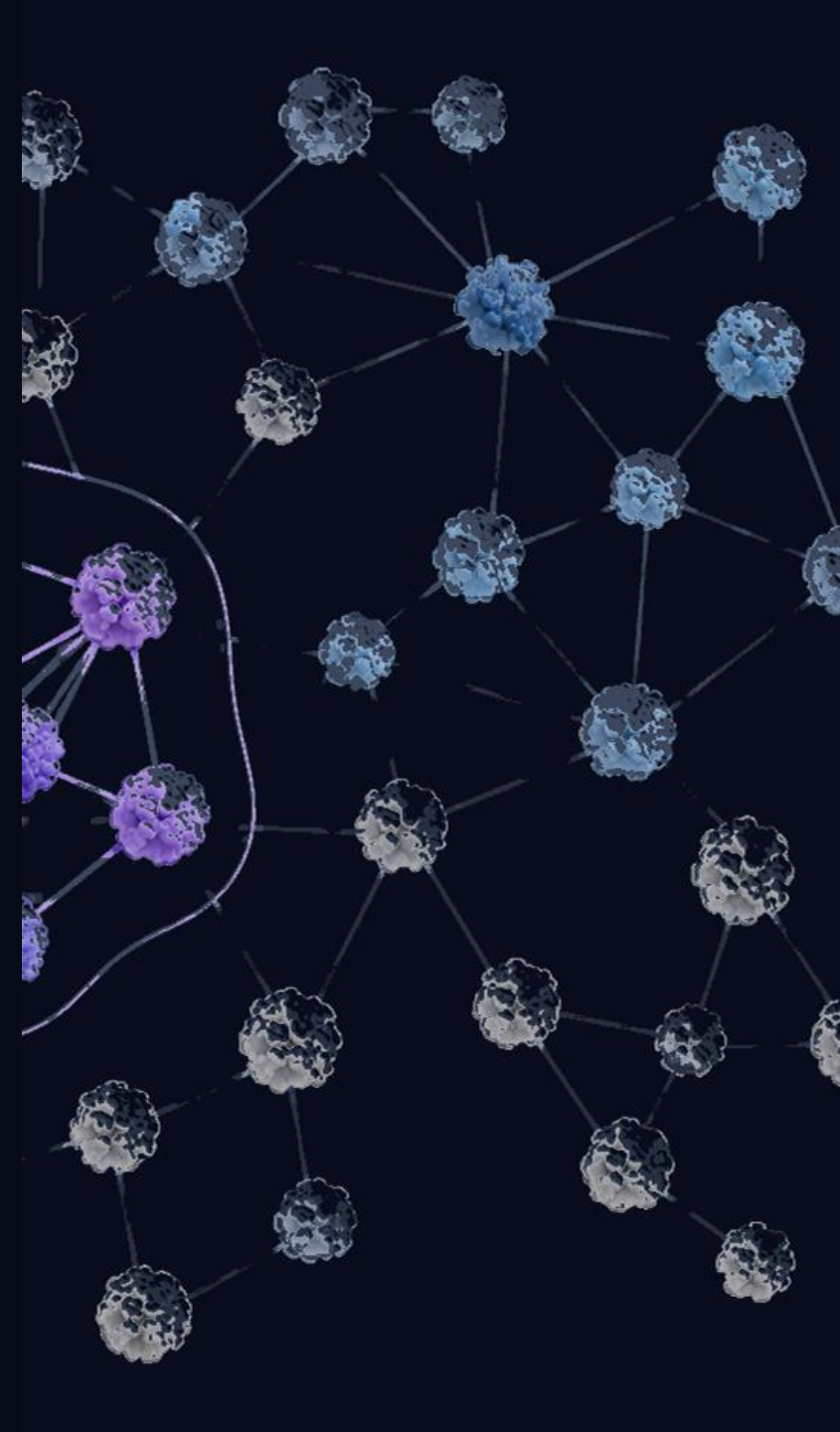
Ask how alterations are distributed across individual tumor samples.

Focus: patterns across patients, not just one percentage

- Do certain genes tend to be altered in the same tumors?
- Are some alterations rarely found together?
- Does one subtype show a more complex pattern of genomic alterations?
- Are mutations and amplifications occurring in the same gene across different tumors?
- Are there distinct molecular subgroups within the selected subtype?

[cBioPortal outputs to use](#)

OncoPrint • Cancer Types Summary • mutation and CNA tracks



Copy-Number & Expression Questions

Ask whether DNA-level changes correspond to altered gene expression.

COPY-NUMBER QUESTIONS

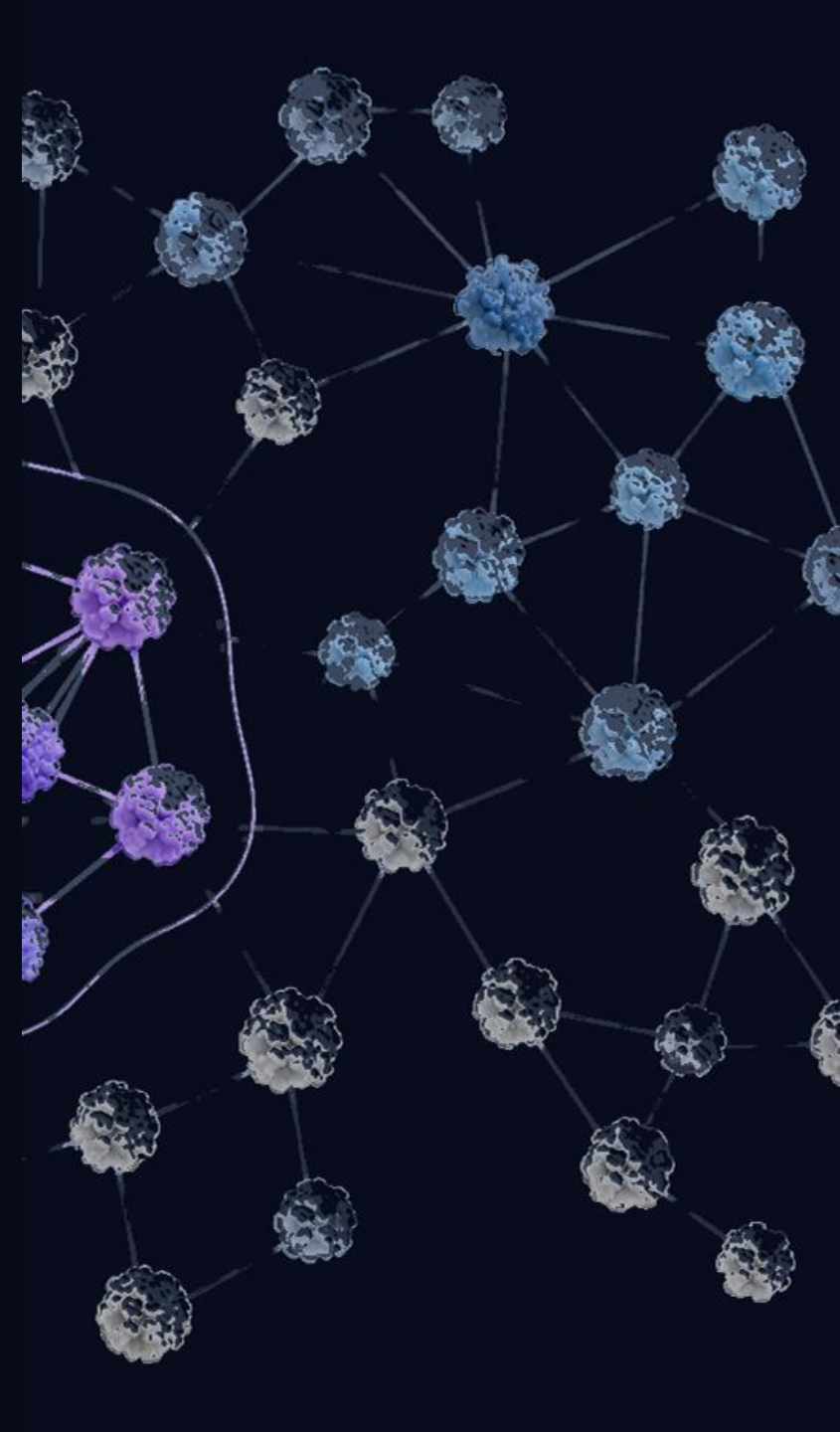
- Is ERBB2 amplification restricted mainly to HER2-enriched tumors?
- Are amplifications more common in one subtype than another?
- Are tumor suppressor genes more often deleted in one group?
- Does a copy-number change correspond to altered mRNA expression?

EXPRESSION QUESTIONS

- Is ESR1 expression higher in Luminal tumors than in Basal-like tumors?
- Is ERBB2 amplification associated with increased ERBB2 expression?
- Do proliferation markers differ between subtypes?
- Does gene expression vary even among tumors assigned to the same subtype?

cBioPortal outputs to use

Plots tab • mRNA tab • Arm-level CNA • OncoPrint

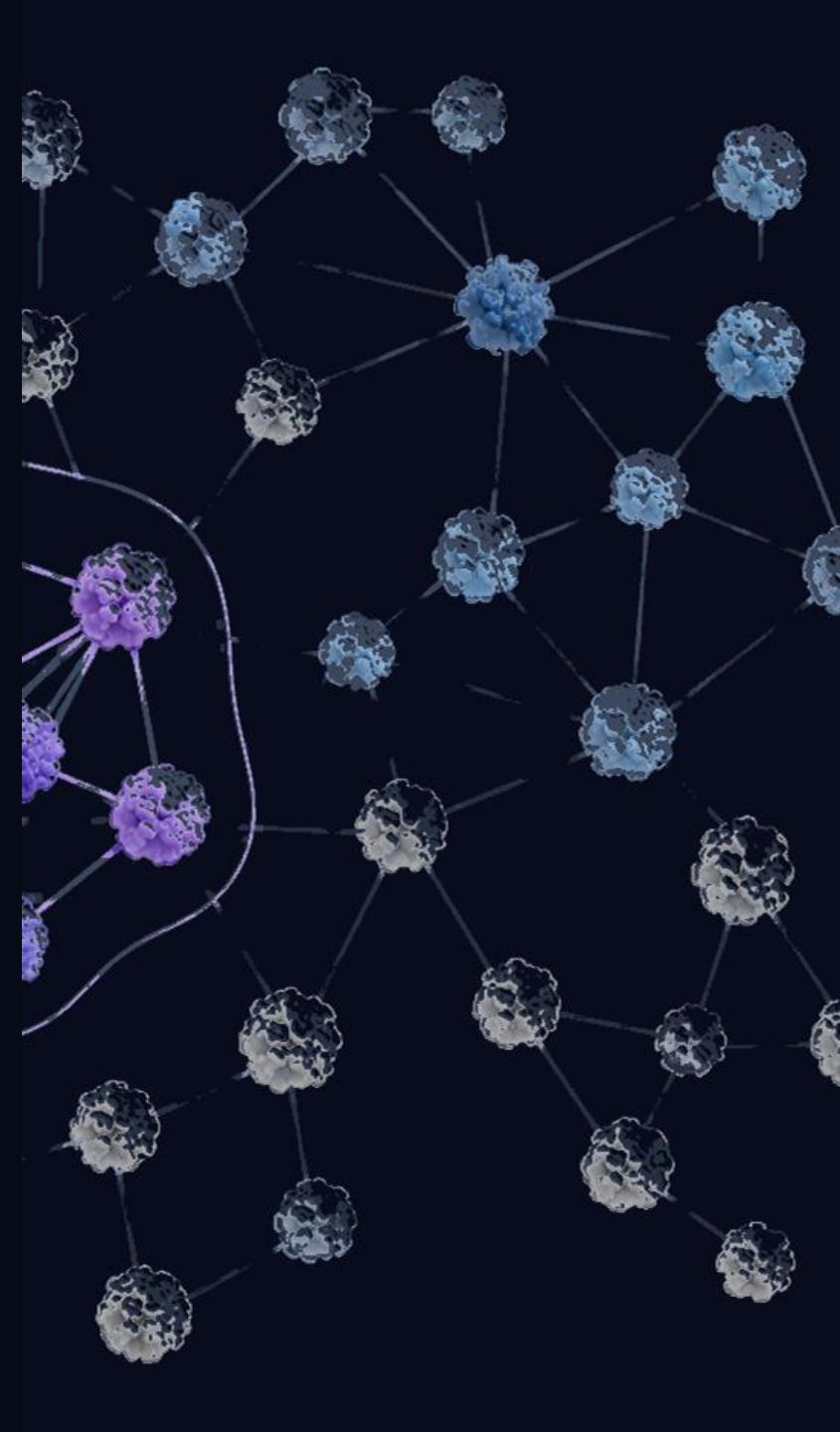


Strong Question Formats

Use these templates to convert a finding into a project question.

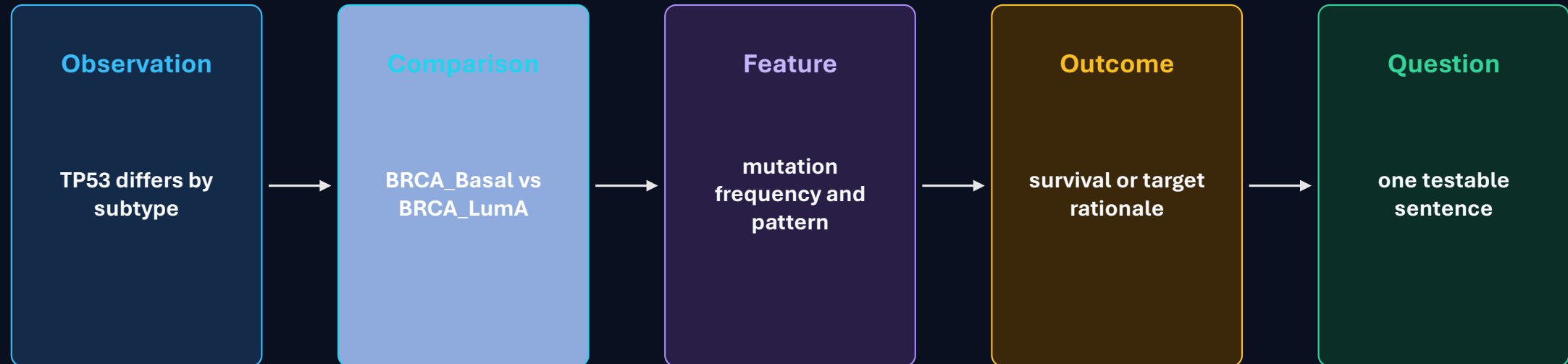
1. Does [gene or alteration] differ between [group A] and [group B]?
2. Is [gene alteration] associated with [survival or another clinical feature]?
3. Do [group A] and [group B] differ in the frequency, type, or location of mutations in [gene]?
4. Is [gene] a plausible subtype-specific target based on its alteration frequency, subtype specificity, and survival association?
5. Are [gene 1] and [gene 2] commonly altered together or separately?

Rule of thumb: one clear comparison + one molecular feature + one interpretable figure.



Turn an observation into a research question

A strong question is narrow enough to be answered using cBioPortal figures.



Formula: In breast cancer, does molecular feature Y differ between group A and group B, and is it associated with biological or clinical evidence Z?

Good research questions are specific

The goal is not to ask the biggest question. It is to ask one that the data can actually answer.

Good

- Do TP53 alterations differ between BRCA_Basal and BRCA_LumA tumors?
- Is PIK3CA alteration enriched in Luminal A tumors?
- Does ERBB2 amplification mark a distinct subtype?
- Are subtype-associated alterations linked to survival outcomes?

Too broad

- What causes breast cancer?
- What is the most important gene in breast cancer?
- How can we cure TNBC?
- Are these mutations bad?

Rule: every question must name the groups, the molecular feature, and the figure that will support the answer.

Project track examples

Keep the project feasible.

Track 1

Subtype comparison

Which genomic alterations distinguish two breast cancer subtypes

- **Subtype chart** to define the groups
- **Genomic Alterations comparison** to identify genes that differ between groups
- **OncoPrint** to visualize alteration patterns across individual tumors

Track 2

Candidate Target Prioritization

Does one gene show evidence for subtype-specific target potential?

- **Genomic Alterations comparison** to assess subtype specificity
- **OncoPrint** to assess alteration frequency and alteration type
- **Survival curve** to examine association with patient outcome

Track 3

Mutation Pattern Analysis

How do mutation frequency, type, and location differ between subtypes?

- **Genomic Alterations comparison** to compare mutation frequency
- **Mutations Beta lollipop plot** to examine mutation type and protein location
- **OncoPrint** to show which tumors carry the mutations

Track 4

Clinical Outcome Association

Is a breast cancer subtype or genomic alteration associated with patient survival?

- **Survival curve** to compare outcomes
- **OncoPrint** to define altered and unaltered groups
- **Genomic Alterations comparison** to identify additional molecular differences between the groups

Core evidence package

Every project needs descriptive evidence, comparative evidence, and interpretation discipline.

1	Cohort context	Subtype chart	How many samples are in each group?
2	Comparative signal	Genomic Alterations	Which genes differ between groups?
3	Gene-level detail	OncoPrint or lollipop	What type of alteration is present?
4	Outcome signal	Survival curve	Is there an association with survival?
5	Limitations	written caveat	What can this analysis not prove?

Target identification vs. validation

You can nominate targets, but they must not overclaim.



Target identification

cBioPortal can identify recurrent, subtype-specific, clinically associated alterations that deserve attention.

Target validation

True validation requires experiments or stronger external evidence showing that changing the target affects cancer behavior.

Correct conclusion language: “supports nomination” or “provides preliminary evidence,” not “proves this is a drug target.”

Starting gene sets

Use small panels. Large panels create crowded figures and shallow interpretation.

Basal-like / TNBC-oriented

TP53 BRCA1 BRCA2 PTEN RB1 MYC EGFR PARP1

Luminal / ER-positive

ESR1 PIK3CA GATA3 FOXA1 CCND1 MAP3K1 CDH1

HER2-oriented

ERBB2 GRB7 TOP2A PIK3CA PTEN AKT1

Cell-cycle / proliferation

TP53 RB1 CCND1 CDK4 CDK6 CDKN2A MYC

You can change genes, but they must be backed up by evidence (e.g. mutation frequency, group differences, etc.)

Figure plan for Week 2

Collect only figures that answer the chosen question.

Required	Subtype chart	Cohort context
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Required	Genomic Alterations	Candidate gene differences
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Required	OncoPrint or lollipop plot	Gene-level alteration pattern
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Conditional	Survival curve	Outcome association
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Optional	Study View plot	Expression, mutation count, or genome alteration comparison
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Naming convention: Date_FigureType_GeneOrComparison.png

Assignments

These are the materials needed before pathway and network interpretation.

1

Finalized research question

5

Survival curve if relevant

2

Comparison groups and gene list

6

Alteration-frequency table for 5–10 genes

3

Genomic Alterations screenshot

7

Three observations and two limitations

4

One gene-level figure

Week 3 will build on this: pathway interpretation, network context, and target-evidence refinement.

A good project is not a pile of figures.

It is one clear question answered with the right evidence.

Choose the question. Then collect only what supports it.

